



**MEHLMANMEDICAL
HY USMLE REVIEW
PART II**



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HY USMLE Review – Part II

- “Hot feels cold; cold feels hot” (temperature dysesthesia) → ciguatera toxicity → toxin blocks sodium channels → caused by consumption of reef fish (mahimahi, Spanish mackerel, etc.)
- Pt with no prior Hx of atopy/asthma + eats meaty fish in sketch location (e.g., Bali) + develops dyspnea + allergic-like reaction → answer = scombroid, not seafood allergy → histidine decarboxylase in decaying fish convert histidine to histamine → allergic-like reaction (often misdiagnosed as allergy)
- Pt gets allergic-like reaction after eating shellfish → answer = shellfish allergy, **not** scombroid (students get all trigger-happy about scombroid after learning about something new, weird, and cool, but if on the USMLE they say shellfish, it’s **shellfish allergy, not scombroid**)
- Vomiting a few hours after eating meat → *S. aureus* preformed heat-stable toxin
- Vomiting (or any unusual Sx like bloody diarrhea) + eating custards, creams, potato salad → answer = *S. aureus* preformed HS toxin → the type of food in this scenario “wins” over the weird bloody diarrhea finding → bear in mind typical bloody-diarrhea-inducing gram (-) rods like EHEC, *Yersinia enterocolitica*, *Campylobacter*, *Shigella*, *Salmonella* have **~1-3-day incubation period**) → iow, if you get sick on the scale of hours from food, *S. aureus* preformed toxin is likely
- Tx of otitis externa → topical ciprofloxacin + hydrocortisone drops
- Prophylaxis for otitis externa (i.e., in someone with continued water exposure, like crew) → topical alcohol-acetic acid drops
- Tx for cerumen buildup → carbamide peroxide drops
- Most common cause of otitis externa → *Pseudomonas*
- Otitis externa + mastoiditis → malignant otitis externa
- Mx of malignant otitis externa → CT or MRI of temporal bone because pus collection is common; if don’t rule out fluid collection and drain it appropriately, can cause brain abscess; this is on one of the pediatric NBME forms, where the answer was CT of the temporal bone in a two-year-old, which is an outrageous dose of radiation for a kid, but it’s the answer on the form; in UW for 2CK, they didn’t list CT, but had MRI and x-ray as answers, and MRI was correct; apparently x-ray is insufficient; mastoiditis will classically present in kid with a pinna that’s displaced upward and outward, often with him or her pulling on it bc of the pain.

- Tx of otitis media → amoxicillin (amoxicillin/clavulanate [Augmentin] is the wrong answer initially; use Augmentin for recurrent OM)
- When to do a tympanostomy tube? → 3+ OM in a 6-month period or 4+ in a year
- Tx for Strep pharyngitis? → Amoxicillin or penicillin only (not Augmentin)
- Aspiration pneumonia or pulmonary abscess; which antibiotic should be given? → clindamycin (anaerobes above the diaphragm)
- Coughs and cups of foul-smelling sputum in COPD, TB, or CF patient → bronchiectasis
- Most common cause of bronchiectasis → worldwide: TB; in western countries: CF
- Young kid + scant white sputum + linear opacity in right-middle lobe on CXR; Dx? → **answer = bronchiectasis caused by right middle lobe syndrome** (no I am not fucking with you; this is on one of the pediatric 2CK forms → 1st question I've ever seen of bronchiectasis where it wasn't cups and cups of foul-smelling sputum; also, if you Google, it, there literally is a peds condition called right middle lobe syndrome that leads to bronchiectasis → search it over some tacos and a Samuel Adams and knock yourself out)
- Chromosome for AR and AD polycystic kidney disease → 6 for AR; 16 for AD
- ADPKD → which do we do for screening, MR angiogram circle of Willis, or serial blood pressure checks? → answer = serial blood pressure checks → don't do MR angiogram screening unless FHx of aneurysm → most patients get high blood pressure from cyst impingement on renal microvasculature → RAAS surges
- Most common extra-renal location for cysts → liver
- Important point about AR: shows up in peds + causes hepatic fibrosis
- Important point about AD: presents in adults; cysts present from birth but just grow + become symptomatic ages 30-40+.
- Empiric Abx therapy for meningitis → ceftriaxone + vancomycin (+/- steroids)
- Lumbar puncture or Abx first in suspected meningitis? → new guidelines say LP first
- When do you do CT head before LP in suspected meningitis?
 - Confusion that interferes with neurologic exam / decreased GCS score
 - Seizure
 - Focal neurologic signs (motor or sensory)

- Papilledema or if the optic fundi cannot be visualized
- Above reasons indicate potential mass lesion, where if you do an LP you can cause tonsillar herniation and death; if CT negative, proceed cautiously to LP
- Bacterial meningitis: low glucose, high protein, high neutrophils (polymorphonuclear cells; PMNs)
- Aseptic (viral) meningitis: normal glucose, normal (or slightly elevated) protein, high lymphocytes
- Fungal meningitis: low glucose, high protein, high lymphocytes (similar to bacterial, but high lymphocytes instead of neutrophils)
- Herpes encephalitis: **lots of RBCs in CSF due to temporal lobe hemorrhage** → CT is often negative, but sometimes Q will mention wave slowing or temporal complexes on EEG
- Difference between meningitis and encephalitis → meningitis is nuchal rigidity (neck stiffness) + photophobia + ophthalmoplegia; encephalitis presents with **confusion**
- Dx of Cryptococcal meningitis? → answer = **latex agglutination** if it's listed over India ink; mucicarmine staining (red stain) can also be done
- Tx for Cryptococcal meningitis → amphotericin B + flucytosine, followed by fluconazole taper
- Nodular density in upper lobe in immunocompromised pt → aspergilloma → next best step = open lung biopsy (sounds radical, but it's the answer on one of the NBME forms) → Tx with -azole → invasive aspergillosis can be treated with caspofungin or voriconazole
- 22M + Hx of three bacterial pneumonias + atopy; presents today with a sore left cheek; Dx? → IgA deficiency (student says "wtf?") → firstly, sore cheek is sinusitis → IgA deficiency is recurrent sinopulmonary infections that "aren't that bad" in a patient "not that young" (that is, it's not a super-sick three-year-old like in SCID or Bruton) → **can present with autoimmune phenomena like atopy and vitiligo**; also can present with Hx of Giardia → apparently Step 1 is also now testing that IgA deficiency means in the Dx of Celiac disease can't do IgA anti-tissue transglutaminase reliably bc clearly the pt wouldn't make the IgA → anaphylaxis after blood transfusion is super HY for IgA deficiency, but also too easy and not mentioned in most Qs.
- Most common immunodeficiency in humans → IgA deficiency
- 17F + 1-year Hx of autoimmune thyroiditis + 2-year Hx of type I DM + Candidal infections since childhood; Dx? → chronic mucocutaneous candidiasis → two points: 1) clearly the Candidal infections can't be due to the diabetes if she's had them since childhood and only DM for two years; 2) USMLE

likes “autoimmune conditions go together,” the same way it likes “autoimmune conditions and immunodeficiencies go together”; in other words, CMC and IgA deficiency are examples of immunodeficiencies with an autoimmune origin, so the vignette mentioning autoimmune phenomenon isn’t an accident.

- Dx of PJPneumonia → bronchoalveolar lavage
- When to add steroids to TMP/SMX for PJP? → A-a gradient >35 or pO₂ <60 mm Hg
- 36M + long-bone fractures + petechiae on the chest → fat embolism
- Motor vehicle accident (MVA) + paradoxical breathing (chest moves outward with exhalation; inward with inhalation) → flail chest
- MVA + rib fractures + underlying infiltrates in lung + low O₂ sats → pulmonary contusion
- MVA + no rib fractures + non-central chest pain + pulmonary infiltrates underlying the painful area → pulmonary contusion (resources will say “white out of the lung” for pulmonary contusion, but this is buzzwordy and never shows up on actual NBME material)
- MVA + pulmonary infiltrates + low O₂ sats + bolus of normal saline given, resulting in worsening of O₂ sats → pulmonary contusion (contused lung is very sensitive to fluid overload)
- MVA + **bruising/pain over the sternum** +/- rib fractures → myocardial contusion
- MVA + bruising/pain over sternum + pulmonary infiltrates + O₂ sats get worse when saline is given → answer = myocardial contusion (“Wait, but I thought you said that latter finding means pulmonary contusion” → it does, and it’s HY for pulmonary contusion, but “bruising/pain over the sternum” wins if it’s listed; this is on a 2CK NBME)
- Important point about Mx of myocardial contusion → do troponins + must monitor for arrhythmia
- Adult male + abdo pain + Hx of alcohol use + diffuse pulmonary infiltrates + low O₂ sats → ARDS
- ARDS must have pO₂/FiO₂ <300
- Tx of ARDS → **low-tidal volume mechanical ventilation** (prevents barotrauma) + increase PEEP
- When you get a random ventilator Q and they want an answer → “increase PEEP” almost always right
- Patient has improving O₂ sats on ventilator; next best step? → “wean from ventilator”
- Thyroid dysfunction on ventilator → euthyroid sick syndrome → normal TSH, normal T₄, low T₃, high reverse T₃ (cortisol decreases peripheral conversion of T₄ to T₃, allowing more to go to reverse T₃)

- Wtf is reverse T3? → an inactive form of T3; all you need to know is that if T4 goes up (e.g., in Graves), reverse T3 should be up because T4 will convert to both T3 and reverse T3; it is also elevated in euthyroid sick syndrome (and T3 down!)
- 25F + intermittent bloody diarrhea for 3 months + intermittent fever + weight loss → answer IBD (Crohn or UC)
- Tx for Crohn + UC → USMLE wants oral sulfasalazine (or mesalamine) first before oral steroids; for perianal disease in Crohn, topical agents / enemas can be used; NBME won't make you pick between oral and topical distinguish (that's more Qbank being pedantic); surgery can be done for UC
- UC + high bilirubin + high ALP → primary sclerosing cholangitis
- Red shins + Crohn → erythema nodosum (type III hypersensitivity; panniculitis → inflammation of subcutaneous fat, not a rash)
- Crater with necrotic debris on forearm + UC → pyoderma gangrenosum
- IBD + eczematoid plaque on forehead + sore joints → psoriatic arthritis (HLA-B27 → PAIR → Psoriasis, Ankylosing spondylitis, IBD, Reactive arthritis)
- IBD + back pain → sacroiliitis (or ankylosing spondylitis)
- Biopsy in Crohn → non-caseating granulomas; transmural; UC you don't see these findings
- Vesicles anywhere on body (not limited for extensors) + Celiac → dermatitis herpetiformis
- Biopsy of DH → IgA deposition at dermal papillae
- Biopsy of small bowel in Celiac → flattening of intestinal villi
- Dx of Celiac → IgA anti-tissue transglutaminase, anti-gliadin (aka anti-endomysial)
- After Celiac Ab positivity, what's the next best step (no further intervention necessary or duodenal biopsy) → answer = duodenal biopsy (sounds wrong, but it's what USMLE wants)
- Weird factoid about Celiac → increased risk of T cell lymphoma
- Biopsy of small bowel in lactose intolerance → normal villi
- Dx of lactose intolerance → hydrogen breath test or decreased stool pH
- Vague vignette where it sounds like either Celiac or lactose intolerance but it's in a young adult who's had zero symptoms until now → lactose intolerance (can be adult-onset)
- PAS-positive macrophages in the lamina propria + arthritis → Whipple disease

- AIDS patient with CD4 count of 47 + confluent ulcers seen on colonoscopy → CMV colitis → confluent means “linear” → CMV = linear ulcers; HSV = punched-out ulcers
- Tx for CMV → ganciclovir (DNA polymerase inhibitor)
- Toxicity of ganciclovir → neutropenia (mouth ulcers + fever)
- Tx of herpes → acyclovir
- Toxicity of acyclovir → nephrotoxicity (crystal nephropathy → acyclovir stones not visible on CT)
- Hereditary angioedema → due to C1 esterase **inhibitor** deficiency
- Tx for HA → danazol (synthetic androgen receptor partial agonist that causes liver to make more C1EI)
- Don't give which drugs to patients with HA → ACEi (can cause angioedema)
- Familial thyroid cancer → medullary (even if they mention nothing else related to MEN 2A/2B); apple-green birefringence on Congo red stain due to amyloid deposition; serum calcitonin high
- Calcitonin mechanism of action → inhibits osteoclast activity (**not** the opposite of PTH; in other words, doesn't put calcium back into bone; it merely caps the Ca that can resorb out of the bone)
- Most common thyroid cancer → papillary; extends lymphatogenously; has papillary structure and psammoma bodies on LM; don't worry about buzzywordy things like Orphan Annie nuclei
- Follicular carcinoma → literally just thyroid follicles on biopsy; will be a cold nodule, like any other type of thyroid cancer (for instance, if you see follicles but it's a hot nodule w/ increased uptake, that's a toxic adenoma, rather than follicular thyroid cancer); spreads hematogenously
- Hashimoto + thyroid cancer + no other histo description → thyroid lymphoma → autoimmune diseases increase the risk of non-Hodgkin lymphoma
- MEN1 → pituitary, pancreas, parathyroid (MEN1 gene; chromosome 11)
- MEN2A → parathyroid, medullary thyroid carcinoma, pheochromocytoma
- MEN2B → medullary thyroid carcinoma, pheochromocytoma, mucosal neuromas, Marfanoid body habitus (“oid” means looks like but ain't)
- Riedel thyroiditis → fibrosis of thyroid → can extend into adjacent structures, e.g., the esophagus, and resemble anaplastic carcinoma
- 52M + abdominal mass + weight loss + biopsy shows lymphocytes + interspersed macrophages → Burkitt lymphoma (student says: “Wtf? I thought Burkitt was the African boy with a jaw lesion.” Yeah,

but in Western countries, Burkitt is usually intra-abdominal; even new studies in Africa have shown a growing preponderance of intra-abdominal Burkitt)

- Translocation for Burkitt → t(8;14), but USMLE also wants you to know t(2;8) and t(8;22)
- Gene for Burkitt → c-myc → transcription factor
- Translocation for CML → t(9;22) → Philadelphia chromosome
- What is the product of the t(9;22) → bcr/abl tyrosine kinase inhibitor
- Tx for CML → imatinib (tyrosine kinase inhibitor)
- Side-effect of imatinib → fluid retention (edema)
- What do you see on bloods in CML → high leukocyte count (mature neutrophils + metamyelocytes + myelocytes)
- Translocation for APL (AML M3) → t(15;17)
- What do you see on blood smear in APL → Auer rods
- What are Auer rods composed of? → myeloperoxidase (cause DIC when released into blood during Tx)
- Translocation for follicular lymphoma → t(14;18)
- Gene for follicular lymphoma → bcl-2 → anti-apoptotic molecule
- Most common indolent non-Hodgkin lymphoma → follicular (waxing and waning neck mass over two years in an adult)
- Most common aggressive NHL → diffuse-large B cell lymphoma (DLBCL)
- Translocation for mantle cell lymphoma → t(11;14)
- 17F + painless lateral neck mass + mediastinal mass; Dx? → Hodgkin lymphoma
- 42M + painless lateral neck mass + hepatomegaly; Dx? → Hodgkin lymphoma
- 40M + Hodgkin lymphoma + renal condition → minimal change disease (“Wtf? Isn’t that kids after viral infection?” → It’s also seen in Hodgkin due to a cytokine effect for whatever magical reason; in UW for 2CK actually)
- Biopsy of lymph node in Hodgkin → Reed-Sternberg cells (“owl eyes” → CD15/30+ B cells)
- Are lymphomas / leukemias normally B or T cell? → almost always B cell

- When is the answer T cell? → When pt has a thymic lesion as evidenced by a positive Pemberton sign (flushing of face with arms above the head) → mediastinal mass in Hodgkin is due to mediastinal lymph node enlargement, not a thymic mass (thymic lesion in Hodgkin exceedingly rare)
- When to give a statin on the USMLE? → [Guidelines from 2019 ACC/AHA](#) → First you need to know important CVD risk factors: **(LDL > 100 mg/dL; hypertension, smoking Hx, CKD, albuminuria, FHx of CVD in first degree relative [sibling or parent])**
 - Anyone age 20-39 with an LDL > 190 mg/dL
 - Anyone age 40-75 with an LDL > 70 mg/dL
 - Age >75 → assessment of risk status + clinician-patient discussion are recommended before commencing or discontinuing a statin
 - Diabetics <40 if one or more above bold risk factor(s) present
 - Anyone age <19 with familial hypercholesterolemia
 - Balance of risk factors contributes to an ASCVD risk score that determines *intensity* of statin administered to the patient (not assessed on USMLE); what the USMLE cares about is you knowing whether or not to give a statin period.
- First drug to start in T2DM → metformin
- Side-effect of metformin → lactic acidosis
- Mechanism of metformin → “increases glycolysis; decreases gluconeogenesis”
- Tx of impetigo → topical mupirocin
- Tx of erysipelas + cellulitis → oral dicloxacillin or oral cephalexin
- Inpatient Tx of MSSA → IV flucloxacillin or IV cephazolin
- Tx of osteomyelitis → IV nafcillin or IV oxacillin or IV cephazolin; add ceftriaxone in sickle cell; add clindamycin in MRSA
- Tx of MRSA skin infection → clindamycin, or TMP/SMX, or doxycycline, or linezolid → vanc has poor skin penetration
- Suspected incarcerated hernia in a vignette → answer = immediate groin exploration
- Most common cause of fever after surgery → atelectasis
- Electrolyte abnormality in Cushing syndrome → hypokalemia (chronically high glucocorticoid can cause potassium wasting distally in the kidney similar to aldosterone)

- Pt has tachy + diaphoresis + diarrhea after drug → serotonin syndrome (tramadol; MOA too soon after stopping SSRI)
- Pt has tachy + diaphoresis + diarrhea + tricuspid valve lesion → carcinoid syndrome
- Cause of carcinoid syndrome → usually small bowel or appendiceal tumor that has metastasized to liver (if not metastasized, liver can process serotonin derivatives it receives); can also be due to bronchogenic carcinoid; tumors are S-100 positive and of neural crest origin
- Dx of serotonin + carcinoid syndromes → urinary 5-hydroxyindole acetic acid (5-HIAA)
- Tx of serotonin syndrome → remove offending agents + administer cyproheptadine (serotonin receptor antagonist)
- Tx of carcinoid syndrome → Tx underlying condition
- Asthma (outpatient) → albuterol (short-acting beta-2 agonist; SABA) inhaler for immediate Mx → if insufficient, start low-dose ICS (inhaled corticosteroid) preventer → if insufficient, maximize dose of ICS preventer → if insufficient, add salmeterol inhaler (long-acting beta-2 agonist; LABA); in other words:
 - **1) SABA; then**
 - **2) low-dose ICS; then**
 - **3) maximize dose ICS; then**
 - **4) LABA.**
- That initial order is universal. Then you need to know last resort is **oral corticosteroids, however they are most effective**. In other words:
- 12M has ongoing wheezing episodes + is on albuterol inhaler; next best step? → add low-dose ICS
- 12M has ongoing wheezing episodes + is on albuterol inhaler; what's most likely to decrease recurrence → oral corticosteroids (student says "wtf? I thought you said ICS was what we do next and that oral steroids are last resort" Yeah, you're right, but they're still **most effective** at decreasing recurrence. This isn't something I'm romanticizing; this distinction is assessed on the FM NBME forms.
- After the LABA and before the oral steroids, any number of agents can be given in any order – i.e., nedocromil or cromolyn sodium, zileuton, montelukast, zafirlukast.
- MOA of nedocromil and cromolyn sodium → mast cell stabilizers

- MOA of zileuton → lipoxygenase inhibitor (enzyme that makes leukotrienes from arachidonic acid)
- MOA of the -lukasts → leukotriene LTC, D, and E4 inhibitors. LTB4 receptor agonism is unrelated and induces neutrophilic chemotaxis (LTB4, IL-8, kallikrein, platelet-activating factor, C5a, bacterial proteins)
- 16M goes snowboarding all day + takes pain reliever for sore muscles afterward + next day develops wheezing out on the slopes again; what's going on? → took **aspirin + this is Samter triad** (now clumsily known as aspirin-exacerbated respiratory disease [AERD]) → triad of aspirin-induced asthma + aspirin hypersensitivity + **nasal polyps**). Just to be clear, other NSAIDs can precipitate Samter triad, but the literature + USMLE will make it explicitly about aspirin.
- 16M takes aspirin + gets wheezing; what are we likely to see on physical exam? → answer on USMLE = nasal polyps.
- "Wait I don't understand. Why would aspirin cause asthma?" → arachidonic acid can be shunted down either the cyclooxygenase or lipoxygenase pathways; if you knock out COX irreversibly by giving aspirin (or reversibly with another NSAID), more arachidonic acid will be shunted down the lipoxygenase pathway → more leukotrienes → more bronchoconstriction
- Kid has Hx of AERD; physician considers agent to decrease his recurrence of Sx → zileuton, or -lukasts (both are correct; and only one will be listed).
- Kid has Hx of AERD; what agent is most likely to decrease his recurrence of Sx → **oral steroids** (sounds wrong, but once again, you need to know oral steroids are most effective for preventing asthma, period; this is exceedingly HY, especially on family medicine forms). We simply don't want to give them because of their nasty side-effects (Cushing syndrome).
- Any weird asthma Tx? → **omalizumab** → monoclonal antibody against IgE → used for intractable, severe asthma unresponsive to oral steroids + in patients who have eosinophilia + high IgE levels (I asked a pulmonologist about this drug years ago when I was in MS3 and he said he was managing 1000 patients with asthma and just three were on omalizumab).
- Acute asthma Mx (emergencies) → most important piece of info straight-up is: USMLE wants you to know that inhaled corticosteroids (ICS) have **no role in acute asthma management**. First thing we do is give oxygen (any USMLE Q that shows depressed O2 sats, answer is always O2) + nebulized

albuterol (face mask with mist); **IV steroids** are then administered. The Mx algorithm is more complicated, but that is what you need for the USMLE.

- Acid-base disturbance in asthma? → respiratory alkalosis → **low O₂, low CO₂, high pH, normal bicarb**
- “Wait, why the low CO₂? Aren’t you not able to breathe?” → low CO₂ is due to high respiratory rate; even if your bronchioles are constricted + filled with secretions, CO₂ can diffuse really quickly; in contrast, O₂ diffuses slowly and requires healthy airways; that’s why with a high RR, O₂ and CO₂ are both low (O₂ can’t get in, but CO₂ can still get out); 19 times out of 20 on the USMLE, if your respiratory rate is high, CO₂ is low.
- “19 times out of 20? Then what’s the exception.” → I’ve seen COPD questions where the patient will have a RR of 28 but a super-high CO₂, and the answer is chronic respiratory acidosis + acute respiratory acidosis (acute on chronic) → in the event of emphysema, where you literally have reduced surface area for gas exchange, even if your RR is high, CO₂ has no way of diffusing out.
- “Wait, why is bicarb normal in acute asthma attack? Shouldn’t it go low to compensate if CO₂ is low?” → not enough time for bicarb to change; takes a minimum of 12-24 hours for renal elimination to have an effect on serum levels; this is why in altitude sickness, where CO₂ is low (due to high RR bc of lower atmospheric O₂), acetazolamide (carbonic anhydrase inhibitor) can be given to increase bicarb loss in the PCT of the kidney to essentially force a metabolic acidosis to compensate.
- 12M + acute asthma episode + given O₂ + nebulized albuterol + IV steroids + his acid-base disturbance is as we talked about above → after 30 minutes, new values are: low O₂, normal CO₂, normal pH, normal bicarb; why? → he’s getting tired → low O₂ means he should still be hyperventilating, so for CO₂ and pH to have normalized means his RR is decreasing → **answer on USMLE = intubate**. When O₂ and CO₂ are both initially down, that’s called a type I respiratory failure; then eventually it will invert, where this patient will have a respiratory acidosis with low O₂, high CO₂, low pH, normal bicarb (type II respiratory failure when O₂ and CO₂ are the opposite).
- Acid-base disturbance in pulmonary embolism? → same as acute asthma: **low O₂, low CO₂, high pH, normal bicarb**.
- Tx of pulmonary embolism → heparin before spiral CT; do V/Q scan in pregnant women.
- In pregnant women, if V/Q scan shows segmental defect; what’s the next best step in Dx? → answer = spiral CT → “Wait, but I thought we don’t do CT in pregnant women.” You’re right. We don’t. But if

for whatever magical reason they ask you what the next best step in Dx is after the V/Q scan is positive, the answer is CT. It doesn't mean we're going to do it. But it's the answer. And this is in UW for 2CK.

- Pt has PE and is already on warfarin → answer = don't give heparin; go straight to spiral CT.
- What about IVC filter? → After the spiral CT confirms the PE, an IVC filter can be inserted in a patient **who's already anticoagulated**. The tricky part is, **don't select IVC filter before doing the CT to confirm**. Students will memorize IVC filter as the answer in someone who's already on warfarin, but think about it: you're not going to stick a fucking filter in someone's IVC before at least confirming the PE with the spiral CT.
- What about tPA in PE? → Never an answer on the USMLE because the indications are debated.
- Is tPA ever an answer for anything? → Ischemic stroke within 4.5 hours. Must do non-contrast CT of the head to confirm no bleed before giving tPA.
- 72M has 30 minutes of facial drooping + arm weakness since waking up from a nap → amount of time since stroke = amount of time that has passed since he was last known to be normal (**i.e., before he went to sleep**) → on the USMLE, this situation often means **don't give the tPA**.
- Regarding contraindications for tPA on the USMLE: most important one is BP, which is 185/110 (if either value is exceeded, don't give tPA). There are many contraindications, but other HY ones are low platelets, high PT or aPTT; Hx of GI bleed past 21 days; Hx of intracranial bleed; recent major surgery.
- 45F + alcoholic + increased creatine kinase → rhabdomyolysis (increased risk in alcoholics)
- 82F + fell in her house + positive for blood on urine dipstick + negative for RBCs on urine LM → rhabdomyolysis (myoglobinuria causes false-positive blood on dipstick)
- Anemia in alcoholism? → non-megaloblastic macrocytic anemia (USMLE will give you high MCV [normal is 80-100] in alcoholic with a bunch of other things going on, and they merely want you to know his high MCV is due to the alcohol)
- Electrolyte abnormality in rhabdo → hyperkalemia (cell lysis + renal failure → myoglobin causes acute tubular necrosis)

- Woman has episiotomy posterior in the midline (vaginal incision made to allow for fetus's head to pass through birth canal without causing vaginal laceration); if the doctor cuts too far, what will he cut into? → **answer = external anal sphincter**
- Stress incontinence → weakened pelvic floor muscles resulting in loss of urine with increased abdominal pressure (coughing, sneezing, laughing) → Hx of multiple pregnancies classic, but often too easy of a descriptor and they won't say that → they'll say there's "downward movement of the vesicourethral junction with coughing"; next best step in Mx? → pelvic floor (Kegel) exercises → if insufficient, do mid-urethral sling; **do not** give medications for stress incontinence (HY!).
- USMLE might ask you which muscle is **not** strengthened by Kegel exercises → student then proceeds to have two thoughts: 1) "wtf, I'm supposed to know Kegel exercises at that high level of detail?" and 2) couldn't *any* muscle not be strengthened by Kegel exercises; I mean, the deltoid wouldn't be for instance." → **answer to this Q = internal anal sphincter** → even if you have zero clue about Kegel exercises, bear in mind **internal sphincters (urethral + anal) are under sympathetic control** → **you can't voluntarily strengthen a muscle not under somatic (voluntary) control**; in case you're curious though, Kegel strengthens levator ani (which comprises pubococcygeus, puborectalis, and iliococcygeus).
- Urge incontinence → answer = "hyperactive detrusor," or "detrusor instability" → needs to run to the bathroom when sticking a key in the front door; needs to run to bathroom when opening car door; answer in multiple sclerosis + menopause (part of vasomotor Sx); can be idiopathic; answer = give oxybutynin (anti-muscarinic) or mirabegron (beta-3 agonist); **once again, do not give these drugs in stress incontinence.**
- Overflow incontinence → answer in diabetes and BPH → neurogenic bladder caused by myelin damage from sorbitol (glucose enters myelin, causing osmotic damage); in BPH, merely due to outlet obstruction → leads to detrusor burnout; **in overflow incontinence, postvoid volume is high (i.e., 300-400 mL in USMLE Qs)**; normal should be <50-75 mL; for diabetic bladder, answer = bethanecol (muscarinic agonist); for BPH, **insert catheter first always.**
- 82M + dribbling, hesitancy, interruption of urinary stream + suprapubic mass (bladder) + bacteria in the urine → answer = insert catheter first, not antibiotics

- 25F + 6 weeks of dysuria + anterior vaginal wall pain + urinary findings normal → answer = chronic interstitial cystitis; must have 6+ weeks of Sx in the absence of any findings; anterior vaginal wall pain sounds weird, but it's on the obgyn NBME forms, and something to bear in mind is that a cystocele (unrelated) will protrude through the anterior superior vaginal wall, so the anatomic proximity is known to cause symptomatology.
- Detrusor muscle → contracts under parasympathetic activity via the pelvic splanchnic nerves (S2-S4); when not voiding, detrusor muscle is kept relaxed by sympathetic activity via the inferior hypogastric nerves (T10-L1).
- Internal urethral sphincter → kept closed by sympathetic activity via inferior hypogastric nerves (T10-L1).
- External urethral sphincter → Opened by somatic activity via the pudendal nerve (S2-4)
- 40M goes hiking + linear vesicles on calf appearing days later → answer = contact dermatitis = type IV hypersensitivity; **linear vesicles** = poison ivy/sumac
- 38F gardens a lot + gets rash on face, arms, backs of hands → answer = contact dermatitis from sunscreen
- Type I HS → **fast-onset; anaphylaxis; atopy (asthma, hay fever [rhinoconjunctivitis], eczema)**; two IgEs bind antigen at Fab fragments + move into close proximity and crosslink → mast cell degranulates + secretes histamine; answer for drug rashes that occur soon after receiving the drug (e.g., sulfa); treat anaphylaxis with IM epinephrine (beta 2 agonism opens the lungs; NE doesn't act on beta 2)
- Type II HS → autoantibodies against one's own cells + receptors
 - **Heparin-induced thrombocytopenia (HIT)**; Abs against platelet factor 4-heparin complex) → treat by stopping heparin + giving direct-thrombin inhibitor (i.e., dabigatran or lepirudin); warfarin is the wrong answer
 - **Graves disease** → activating TSH Abs against TSH receptor
 - **Goodpasture syndrome** → "2, 3, 4... 2, 3, 4... 2, 3, 4. The Goodpasture is marching in the field, 2, 3, 4!" → Type 2 hypersensitivity against the alpha-3 chains of type 4 collagen
 - **Pernicious anemia** → Abs against intrinsic factor or parietal cells → decreased B12 absorption through terminal ileum

- **Warm autoimmune hemolytic anemia** → Coomb positive, meaning IgG targets RBCs → seen in things like ABO mismatch, drugs/infection; CLL
- **Cold autoimmune hemolytic anemia** → IgM Abs against RBCs → classically Mycoplasma
- **ITP (immune thrombocytopenia purpura)** → Abs against GpIIb/IIIa on platelets → low platelet count + high bleeding time → treat with steroids, then IVIG, then splenectomy
- Type III HS → Ab-Ag complexes
 - **Serum sickness** → immune complexes depositing in joints several days after drug or infection
 - **Arthus reaction** → localized cutaneous type III HS → skin reaction/rash at injection site several days after administration of agent → key point is that it's not immediate; in type I hypersensitivity, we'd get an immediate reaction; in Arthus reaction, it takes a few days
 - Super HY is **postinfectious arthritis** → arthritis due to HepB, HepC, rubella, Yersinia enterocolitica, etc. → immune complex deposition in joints → this is considered a serum sickness
 - **Erythema nodosum** → panniculitis (inflammation of subcutaneous fat) due to immune complex deposition → USMLE likes **Crohn disease**, sulfa drugs, Strep infections, and Coccidioidomycosis as causes.
 - **Post-streptococcal glomerulonephritis (PSGN)** → can occur after skin infections as well (i.e., impetigo, erysipelas, cellulitis)
- Type IV hypersensitivity → T cell-mediated response (delayed)
 - Contact dermatitis (nickel [wristwatches], sunscreen, poison ivy/sumac, rubber)
 - PPD test for TB
 - Multiple sclerosis (T cell attack against oligodendrocytes in white matter)
 - Stevens-Johnson syndrome; toxic epidermal necrolysis
- Need to know low bicarb in pt with dehydration (or any type of shock) → answer = lactic acidosis → decreased perfusion to vital organs → decreased oxygen delivery → increased anaerobic respiration → increased lactic acid
- High-anion gap metabolic acidosis → MUDPILES → methanol, uremia, DKA, phenformin, iron/INH, lactic acidosis, ethylene glycol, salicylates

- Tx for ethylene glycol and methanol toxicity → fomepizole **only** if listed → in other words, if you get combinations with “ethanol first then fomepizole,” etc., just choose **“fomepizole only”** if it’s listed
- Paint thinner + blurry vision → methanol toxicity (on one of the later Step 1 NBMEs)
- Which decreases risk of stroke more, smoking cessation or BP control? → BP control (USMLE wants lisinopril as the answer, not smoking cessation); for stroke, answer is always hypertension as worse risk factor
- High HbA2 → answer = beta-thalassemia
- Dx of alpha or beta thalassemia → hemoglobin electrophoresis
- Pregnant woman has low serum iron that doesn’t improve with iron supplementation → answer = do hemoglobin electrophoresis → Dx = thalassemia (usually alpha trait with one mutation because that’s asymptomatic)
- Dx of sickle cell → hemoglobin electrophoresis
- Dx of multiple myeloma → serum protein electrophoresis, then bone marrow biopsy
- Adult male + works in manufacturing/factory + cognitive decline + microcytic anemia → answer = lead poisoning → inhibits delta-ALA dehydratase + ferrochelatase → **microcytic anemia is really HY for lead poisoning** → will often see basophilic stippling or RBCs
- What kind of RBCs in thalassemia → target cells
- Infection + RBCs lacking central pallor + positive Coomb test → answer = hemolytic anemia, **not** hereditary spherocytosis → spherocytes are seen in drug- and infection-induced hemolytic anemia, not just in HS → difference is Coomb (IgG against RBCs) is positive in Ab-induced hemolytic anemia, but clearly not in HS bc the latter is cytoskeletal (ankyrin, spectrin, band protein deficiency) in etiology.
- 12M + red urine 1-3 days after upper respiratory tract infection (URTI) → IgA nephropathy, not PSGN; can also get IgA nephropathy from GI infections
- 12M + red urine 1-2 weeks after URTI or skin infection → PSGN → can get it from Group A Strep skin infections
- 6F + red urine + abdo pain + arthralgias + violaceous lesions on buttocks + thighs; Dx? → Henoch-Schonlein purpura; red urine = IgA nephropathy → HSP is tetrad of 1) IgA nephropathy, 2) palpable purpura, 3) arthralgias, 4) abdo pain

- 23F + unilateral resting tremor + hemolytic anemia + increased LFTs; next best step in Mx? → **answer = do a slit-lamp exam** → Dx = Wilson disease; Parkinsonism in a young patient = Wilson disease until proven otherwise → low serum ceruloplasmin + increased urinary copper
- How is copper normally excreted by the body? → through bile; defective excretory pump in Wilson
- Tx for Wilson disease → penicillamine
- Tx for hereditary hemochromatosis → serial phlebotomy
- Tx for secondary hemochromatosis (transfusional siderosis) due to chronic blood transfusions → chelation therapy
- Pink line tracking up the arm from the palm in 35M who burned himself recently; Dx? → lymphangitis → sporotrichosis is wrong answer (can cause lymphatogenous spread)
- Brown recluse spider bite → abdominal pain; manage symptomatically
- Black widow spider bite → causes hypocalcemia → give IV calcium gluconate
- Woman who works in dental care with vesicle on finger; Dx? → herpetic whitlow → HSV1/2 of the finger → treat with oral acyclovir (in real life will give both oral and topical; on the USMLE, answer is always oral)
- Suspected skin cancer in non-cosmetically sensitive area → surgical excision
- Suspected skin cancer on face → Mohs surgery (thin slices looking for positive margins)
- Suspected skin cancer on neck → full-thickness biopsy
- Most important prognostic factor for melanoma → **depth** of lesion
- Recommendation to prevent melanoma → avoid sun or use protective clothing (sounds absurdly obvious, but I've seen students choose wrong answers like use SPF30)
- Farmer/fisherman/construction worker + red, scaly lesion on ear/forehead; Dx? → actinic keratosis
- Above patient with ulcerated lesion emerging from the red scaly lesion; Dx? → SCC → need to know actinic keratosis is precursor to SCC
- Risk factor for SCC apart from the sun → **USMLE is obsessed with immunodeficiency and smoking, even for cutaneous SCC**
- Biopsy of SCC? → keratin pearls + intercellular bridges

- Old scar or burn + new ulcerated lesion → answer = SCC (Marjolin ulcer) → chronically irritated area / burn / scar can lead to SCC (when I was in MS4 I saw an older woman with a Marjolin ulcer on her chin from a chickenpox scar she had since she was a kid)
- Ulcerated lesion on nose or pinna of ear + rolled/heaped-up edges → BCC
- Ulcerated lesion + telangiectasia visible → BCC
- Biopsy of BCC? → islands and nests of basophilic cells (dark purple)
- Blood transfusion + → fever + negative Coombs test; Dx? → febrile non-hemolytic transfusion reaction
- Most common blood transfusion reaction? → febrile non-hemolytic transfusion reaction
- Tx for febrile non-hemolytic transfusion reaction → **acetaminophen on the NBME, not prednisone**
- What causes febrile non-hemolytic transfusion reaction? → Abs against donor MHC antigens on RBCs or cytokines from leukocytes in the donor blood
- Blood transfusion + fever + chills + flank pain + hypotension → ABO mismatch (and positive Coombs)
Decreased hemoglobin + increased bilirubin 2-4 weeks after blood transfusion; Dx? → delayed transfusion reaction → caused by the presence of recipient “amnesic” antibodies (weird, but just memorize it)
- Blood transfusion + dyspnea + hypoxemia + bilateral pulmonary infiltrates; Dx? → TRALI (transfusion related acute lung injury)
- What causes TRALI? → donor Abs against recipient MHCs → activated neutrophils cause alveolar damage → most common cause of transfusion-associated death
- Dude moves into new apartment building and uses the hot tub a lot + has low oxygen sats + dyspnea + crackles at lung bases + no wheezing + low-grade fever → answer = **hot tub lung** (outrageous, but an answer on an NBME exam) → caused by hot tube fumes; not related to Pseudomonas (hot tub folliculitis) (weird factoid to end this pdf but we’ll run with it)



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MEHLMANMEDICAL HY USMLE REVIEW PART II

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